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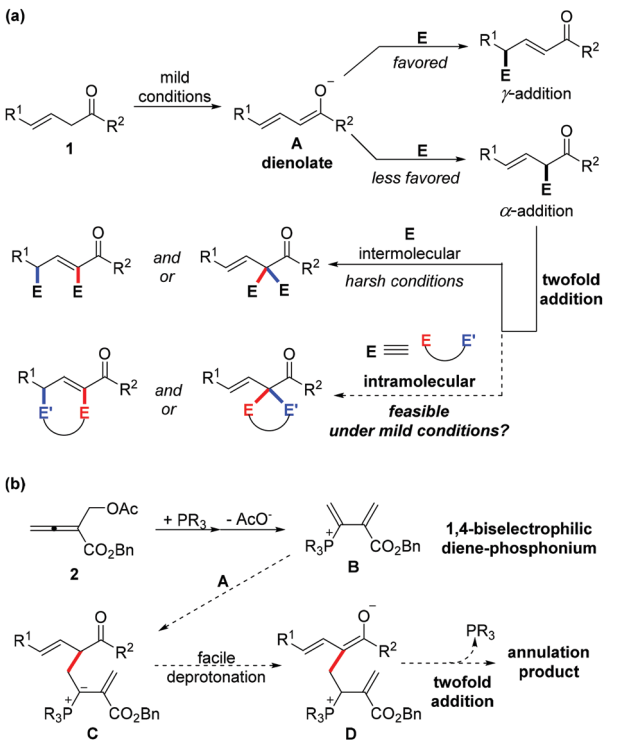
Phosphine-catalysed (4+1) annulations of β′-acetoxy allenoate with β,γ-unsaturated carbonyl compounds†

Yueqi Zhang, Danfeng Wang and Xiaofeng Tong *

While β,γ-unsaturated carbonyl compounds have been widely used as γC- or αC-nucleophiles, their potential αC,αC-bisnucleophilic reactivity is still underdeveloped. Herein, a phosphine-catalysed (4+1) annulation of β′-acetoxy allenoate and a β,γ-unsaturated carbonyl compound is reported, wherein β′-acetoxy allenoate is the 1,4-biselectrophilic component while the β,γ-unsaturated carbonyl compound serves as an αC,αC-bisnucleophile. The process not only provides a new reaction mode of β,γ-unsaturated carbonyl compounds under mild conditions but also broadens the scope of Lewis base-catalysed annulation of acetoxy allenoate.

The addition of dienolate to various electrophiles is a useful process to forge C–C and C–N bonds.[1] In this context, β,γ-unsaturated carbonyl compounds **1** are particularly attractive since the presence of such two functional groups renders the αC–H higher activity,[2] being amenable to generate dienolate species **A** under mild conditions (Scheme 1a). For such species, two divergent addition pathways have been achieved, one is the vinylogous nucleophilic addition of γC to various electrophiles, such as carbonyl compounds, imines and Michael acceptors;[3] the other is the addition to αC albeit with relatively less competitiveness, which may be ascribed to the fact that switching the reactive site from the γC to the αC requires the disruption of the π-conjugation system inherent to the dienolate.[4] Inspired by these well-known reports, it is logical to postulate that the specific β,γ-unsaturated carbonyl structure remaining in the resulting α-addition product would allow for further addition. However, this proposed twofold addition would become, if possible, extremely difficult likely due to the increased steric hindrance and thus harsh conditions might be unavoidable (Scheme 1a).[5] We envisioned that utility of a highly reactive 1,*n*-biselectrophile might impart the second addition much higher efficiency in the intramolecular fashion, achieving a novel (3+*n*) and/or (1+*n*) annulation protocol between the β,γ-unsaturated carbonyl compound and 1,*n*-biselectrophile under mild conditions (Scheme 1a).

Our group realized the phosphine-catalysed annulation of β′-acetoxy allenoate **2** in 2010.[6] Since then, the annulations of actoxy allenoates under Lewis base catalysis have been extensively studied by several groups.[7] The key to these successes is the involvement of 1,4-biselectrophilic diene-phosphonium species **B** (Scheme 1b),[6,7] which is in sharp contrast to traditional phosphine-catalysed annulations relying on the phosphonium-carbanion zwitterion.[8] With these considerations in mind, we



Scheme 1 Overview of reaction modes of β,γ-unsaturated carbonyl compounds and the design plan of this work.

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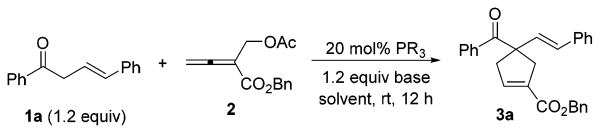
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hypothesized that a Michael-type addition of **A** to **B** would be feasible to generate zwitterion **C**. More importantly, this *in situ* generated species **C** might be able to serve as a base to enforce the generation of dienolate **D**, which was able to trigger a second intramolecular addition to furnish an annulation process. Herein, we report the phosphine-catalysed (4+1) annulation of **1** and **2**, which would not only provide a new reaction mode of β,γ-unsaturated carbonyl compounds under mild conditions but also broaden the scope of Lewis base-catalysed annulation of allenoate **2** (Scheme 1b).

To verify our hypothesis, we conducted the reaction of β,γ-unsaturated carbonyl compound **1a** and allenoate **2** in toluene at room temperature with the use of PPh_3 (20 mol%) as a catalyst and Na_2CO_3 as a base additive, which indeed afforded a (4+1) annulation product **3a** albeit only in 30% yield (Table 1, entry 1). When electron-richer phosphine catalysts, such as PPh_2Me and $PPhMe_2$, were instead employed under the otherwise identical conditions, product **3a** was isolated in 55% and 41% yield, respectively (Table 1, entries 2 and 3). These results implied that the electronic character of phosphine played an important role in the reaction efficiency. While much more electron-rich P^nBu_3 gave product **3a** in 28% yield, the use of P(*p*-MeO–C_6H_4)$_3$ improved the yield to 71% (Table 1, entries 4 and 5). As expected, either less electron-rich P(*p*-Me–C_6H_4)$_3$ or electron-poor P(*p*-F–C_6H_4)$_3$ was found to be inferior to P(*p*-MeO–C_6H_4)$_3$ (Table 1, entries 6 and 7). Unfortunately, we failed to further improve the reaction efficiency when other reaction parameters, such as base additive, solvent and temperature, were extensively examined (see the ESI,† for details).

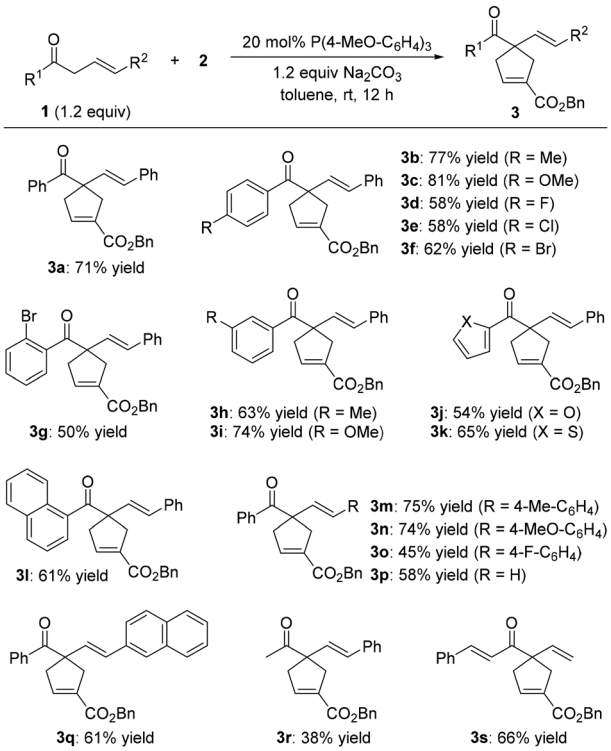
With the optimized reaction conditions being established, we then explored the substrate scope of the (4+1) annulation and the results are summarized in Scheme 2. In general, these reactions took place smoothly under the standard conditions and afforded cyclopentene products in moderate to good yields. A wide range of 1,4-diphenyl-3-butenones **1a–1f** with different electron-bias phenyl ketone moieties were well tolerated.

Table 1 Optimization of the reaction conditions[a]



Entry	PR_3	Base	Solvent	Yield (%)[b]
1	PPh_3	Na_2CO_3	Toluene	30
2	PPh_2Me	Na_2CO_3	Toluene	55
3	$PPhMe_2$	Na_2CO_3	Toluene	41
4	P^nBu_3	Na_2CO_3	Toluene	28
4	P(4-MeO–C_6H_4)$_3$	Na_2CO_3	Toluene	71
5	P(4-Me–C_6H_4)$_3$	Na_2CO_3	Toluene	60
6	P(4-F–C_6H_4)$_3$	Na_2CO_3	Toluene	22
7	P(4-MeO–C_6H_4)$_3$	Cs_2CO_3	Toluene	34
8	P(4-MeO–C_6H_4)$_3$	NEt_3	Toluene	56
9	P(4-MeO–C_6H_4)$_3$	Na_2CO_3	THF	47

[a] Reaction conditions: substrate **1a** (0.24 mmol, 1.2 equiv.), allenoate **2** (0.2 mmol), phosphine catalyst PR_3 (0.04 mmol), base (0.24 mmol, 1.2 equiv.), solvent (3 mL), room temperature. [b] Isolated yield.



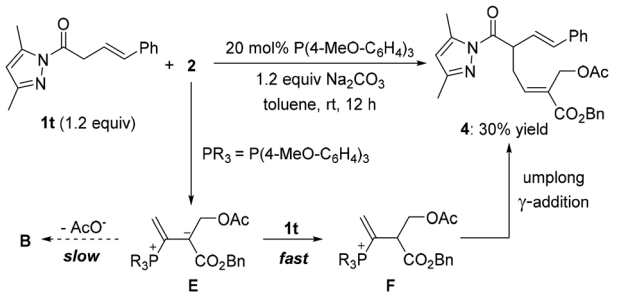
Scheme 2 The reaction scope of (4+1) annulation.

Among them, substrates **1b** and **1c** bearing an electron-rich phenyl group gave the corresponding products in relatively higher yields than the ones with an electron-poor phenyl group including substrates **1d–1f**. These results indicated that the rate-determining step might not involve the generation of dienoate species. The yields of products **3g** and **3l** respectively dropped to 50% and 61% likely due to the increased steric hindrance. The reaction was also applicable to the substrates bearing a heteroaryl ketone moiety, such as furan and thiophene. The corresponding products **3j** and **3k** were isolated in moderate yields. Then, the effect of substituent on the alkene moiety was briefly explored. Again, substrates **1m** and **1n** bearing an electron-rich phenyl group exhibited somewhat higher reactivity than substrate **1o**. 1-Phenyl-3-butenone **1p** was also a suitable substrate to deliver product **3p** in 58% yield. In contrast to aryl ketone substrates **1a–1q**, methyl ketone substrate **1r** showed much less reactivity, affording product **3r** only in 38% yield. Notably, 1-phenyl-1,5-hexadien-3-one **1s** readily reacted with allenoate **2**, giving product **3s** with three distinct alkenes.

When the protocol was applied to β,γ-unsaturated *N*-acylpyrazole **1t**,[9] it was surprising to isolate compound **4** in 30% yield while the expected (4+1) annulation product was not observed at all (Scheme 3).[10] Likely due to its much more acidic αH, *N*-acylpyrazole **1t** could be rapidly deprotonated by zwitterion intermediate **E** that was resulted from the addition of phosphine to allenoate **2**. Vinyl phosphonium intermediate **F** was thus generated, which was responsible for the formation of product **4** *via* an umpolung γ-addition mechanism[11] (Scheme 3).

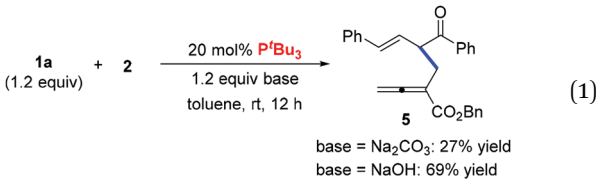
These observations are against the previous reports wherein intermediate **E** is generally believed to rapidly undergo

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Scheme 3 The phosphine-catalysed reaction of β,γ-unsaturated *N*-acylpyrazole **1t** with allenoate **2**.

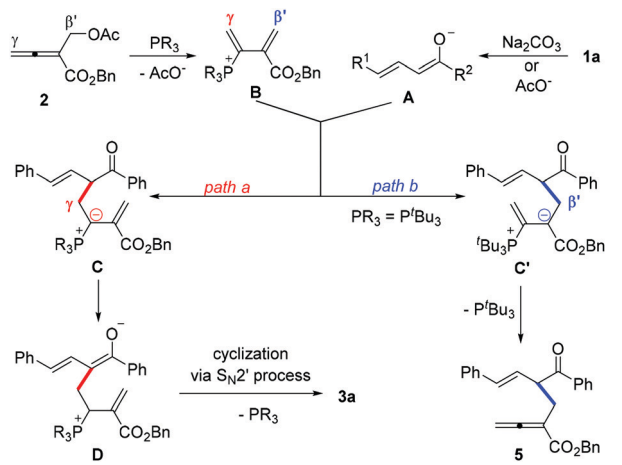
1,2-elimination of acetoxyl to form diene-phosphonium **B**. Although this unexpected result closely related to the unique β,γ-unsaturated *N*-acylpyrazole substrate may be a specific case, it indeed provides a good clue to re-consider the chemical behaviour of allenoate **2** under Lewis base catalysis.[10]



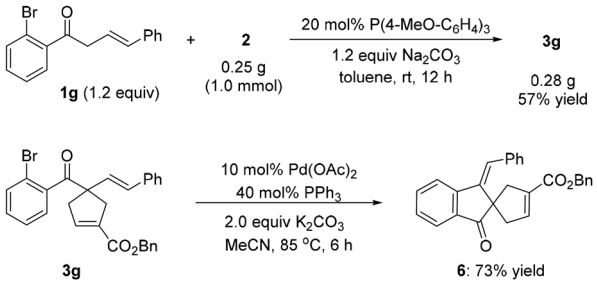
(1)

More interestingly, the reaction of **1a** and **2** with the use of P^tBu_3 as a catalyst under the otherwise identical conditions instead produced compound **5**, clearly pointing out that a reaction mechanism different from the one leading to cyclopentene would be involved (eqn (1) and Scheme 4).[12]

Taking these results into account, a plausible mechanism for the (4+1) annulation of **1a** and **2** is depicted in Scheme 4. Our previous studies supposed that both the γ- and β′-positions are available for nucleophilic attack, which lead to zwitterionic intermediates **C** (path a) and **C′** (path b),[12] respectively. Then, the resulting dienoate **D** undergoes an intramolecular S_N2'-type process to afford product **3a** and regenerate the phosphine catalyst (path a). On the other hand, intermediate **C′** may undergo 1,2-elimination of the phosphine catalyst to produce



Scheme 4 A plausible mechanism.



Scheme 5 Scale-up reaction and synthetic transformation.

compound **5** (Scheme 4). We believed that the γ-position of intermediate **B** might become congested due to the presence of a P^tBu_3 group, thus switching the attack of dienoate **A** from the γ-position to the β′-position to form **C′** with high regioselectivity (Scheme 4).

To check the practicality of the process, the reaction of **1g** and **2** was carried out in 2.0 mmol scale under the optimal conditions, which gave product **3g** in a synthetically useful yield (57%). Moreover, product **3g** could be converted into spiro[4.4]nonane derivative **6** in 73% yield by using common Heck reaction conditions, highlighting the applicability of this protocol toward the synthesis of a structurally complex framework (Scheme 5).

In summary, we have developed the phosphine-catalysed (4+1) annulation of β′-acetoxy allenoate and a β,γ-unsaturated carbonyl compound, which provides facile access to cyclopentenes of synthetic value under mild conditions. In this process, the β,γ-unsaturated carbonyl compound serves as a αC, αC-bisnucleophile and nicely matches the reactivity of the involved diene-phosphonium species to accomplish (4+1) annulation, which complements the well-known vinylogous γ-addition and α-addition of a β,γ-unsaturated carbonyl compound. Further investigation on the asymmetric version of the (4+1) annulation is underway and will be reported in due course.

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Conflicts of interest

There are no conflicts to declare.

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